AARAV DESHMUKH
Mechanical Design Engineer | SOLIDWORKS | FEA | ANSYS Mechanical | GD&T | CFD
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https://linkedin.com/in/aarav-demo | https://github.com/aarav-demo/engineering-portfolio

SUMMARY
Mechanical engineering graduate with dated internship and competition hardware experience. Propulsion, structures & flight hardware.

EDUCATION
Pune Institute of Technology | B.Tech in Mechanical Engineering | Jun 2026
CGPA: 8.6 / 10.0

INTERNSHIPS
Orbital Works - Mechanical Design Intern | Bengaluru, Karnataka | Jan 2026 - Jul 2026
- Designed cryogenic feed systems in SOLIDWORKS with Geometric Dimensioning and Tolerancing (GD&T), reducing valve assembly mass by 18%.
- Validated a propulsion systems bracket using Finite Element Analysis (FEA) in ANSYS Mechanical, achieving a Factor of Safety (FOS) of 1.85 under 4g load.
- Simulated regenerative cooling nozzle channels using Computational Fluid Dynamics (CFD) in ANSYS Fluent, reducing pressure drop by 14 kPa.

ENGINEERING PROJECTS
Formula Student Team - Structures Lead | Pune, Maharashtra | Aug 2025 - Dec 2025
- Optimized a Carbon Fiber Reinforced Polymer (CFRP) monocoque in SOLIDWORKS using Design for Manufacturability (DFM), reducing frame mass by 4.2 kg.
- Fabricated a propulsion systems test fixture in SOLIDWORKS, achieving a machining tolerance of 0.05 mm.

TECHNICAL SKILLS
SOLIDWORKS | Finite Element Analysis (FEA) | ANSYS Mechanical | Geometric Dimensioning and Tolerancing (GD&T) | Computational Fluid Dynamics (CFD)

CERTIFICATIONS
Engineering design and hardware safety workshop | Feb 2026